Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the accuracy during training.

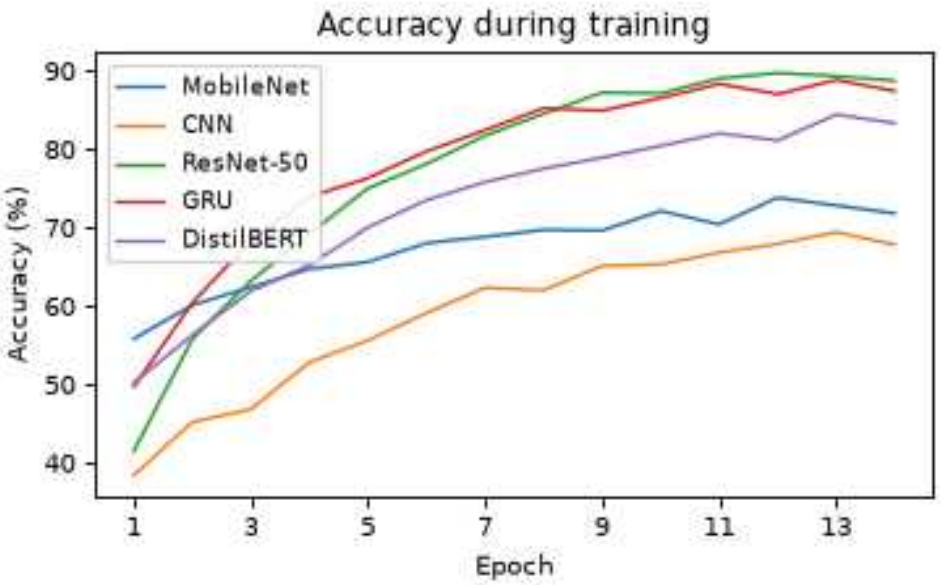


Figure 1: Accuracy during training for the models in this experiment.

ResNet-50 reaches 60.4% accuracy at epoch 2 (Figure 1). As shown in Figure 1, CNN reaches about 70% accuracy at epoch 13. CNN reaches 45.2% accuracy at epoch 2. At epoch 6, ResNet-50 is 15.9 points better than DistilBERT.

Figure 2 shows the accuracy of BERT-base by dataset.

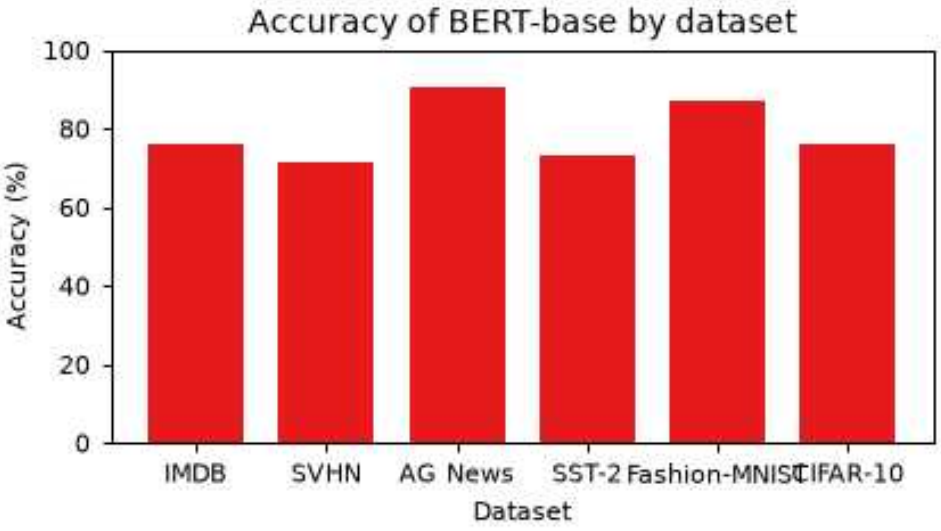


Figure 2: Accuracy of BERT-base by dataset for the models in this experiment.

BERT-base scores 4.0 points higher on AG News than on IMDB. As shown in Figure 2, BERT-base scores 67.1 points higher on AG News than on SST-2. As shown in Figure 2, BERT-base achieves 87% accuracy on Fashion-MNIST. BERT-base reaches 71.3% accuracy on SVHN.

Figure 3 shows the test accuracy of ResNet-18 by dataset.

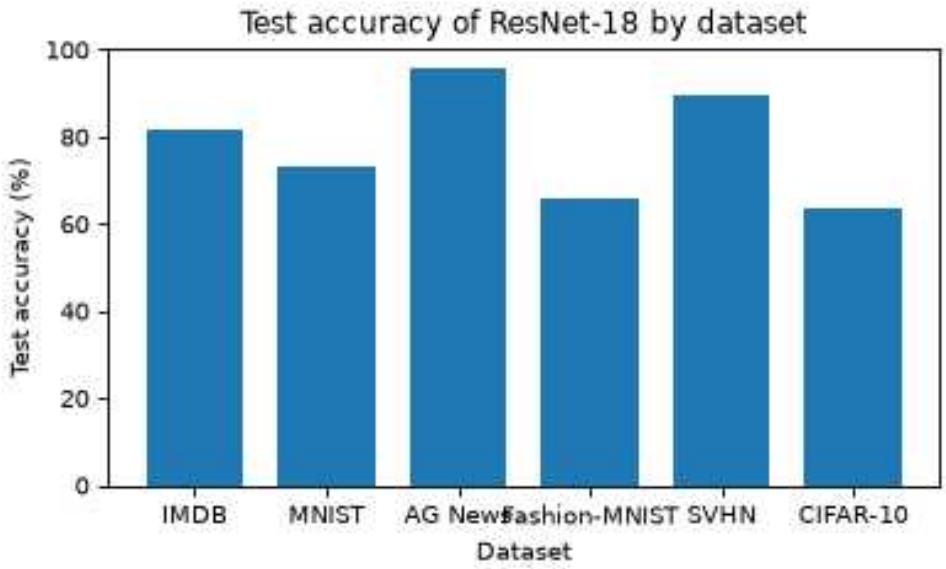


Figure 3: Test accuracy of ResNet-18 by dataset for the models in this experiment.

As shown in Figure 3, ResNet-18 scores 55.8 points higher on AG News than on SVHN. ResNet-18 reaches 73.0% test accuracy on MNIST (Figure 3).

Figure 4 shows the validation accuracy by dataset.

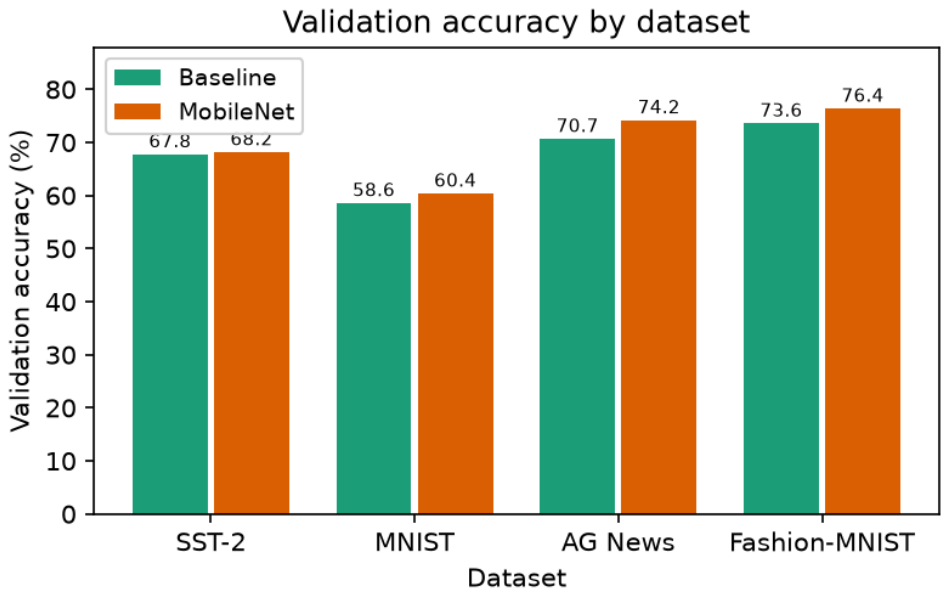


Figure 4: Validation accuracy by dataset for the models in this experiment.

On SST-2, MobileNet obtains a validation accuracy of 67.8%. On Fashion-MNIST, MobileNet is 2.8 points better than Baseline (Figure 4).

3 Discussion

XGBoost reaches 50.2% accuracy on AG News. We note that the training was stable for all models.